Detailed Archival Research Dossier: Frederic Middlebrook Richards (1925–2009)

The following research dossier provides exhaustive raw material, evidentiary analysis, and primary source documentation regarding the life and scientific contributions of Frederic Middlebrook Richards. This document is intended to serve as a comprehensive resource for the production of a biographical narrative for the BIOL 3030 introductory biochemistry course. The materials are organized to highlight the intersection of Richards' "Yankee engineering" ethos with the mid-20th-century revolution in structural biology.

A) ANCHOR SCENES

SCENE #1: The Midnight Restoration (The RNase S Titration)

Summary

This scene captures the precise moment when the physical nature of the protein "fold" was demonstrated through functional restoration. Having previously used the protease subtilisin to cleave Ribonuclease A into two fragments—the S-peptide and the S-protein—Richards sought to determine if these pieces, which were individually inactive and held together by no covalent bonds, could reconstitute a functional enzyme simply by being reunited in solution.

Date/Place December 7, 1957, at approximately 7:00 PM; Richards' laboratory at Yale University, New Haven, Connecticut.

People Frederic Richards (working alone); the distant influence of Christian Anfinsen (who provided the initial samples) and Kaj Linderstrøm-Lang (the source of the subtilisin method).

Concrete Details The setting was a quiet laboratory during the evening hours, a time Richards often utilized for focused bench work. The central instrument was a Beckman-DU spectrophotometer, a staple of 1950s biochemistry. Richards had two separate vials: one containing the S-peptide (the first 20 amino acids of the chain) and one containing the S-protein (the remaining 104 amino acids). Individually, when added to a solution of RNA, the spectrophotometer needle remained still, indicating zero enzymatic activity. Richards began a titration, slowly adding the S-peptide to the S-protein while monitoring the absorbance. He observed the needle of the Beckman-DU climb steadily, tracking a linear increase in activity that reached exactly 100% when the molar ratio reached 1:1. This restoration occurred despite the absence of any chemical reagent to re-form the peptide bond; the fragments simply "clicked" into place through non-covalent interactions.

Why it Matters This experiment provided the first functional proof that the three-dimensional structure of a protein is an inherent property of its amino acid sequence and is stabilized by

non-covalent forces (hydrophobic interactions and hydrogen bonding). It served as the experimental foundation for the "thermodynamic hypothesis" of protein folding that would later earn Christian Anfinsen the Nobel Prize. For Richards personally, he described this as the "high point of my career in terms of excitement".

Evidence Quality Primary. Richards recounted the specific date and time in his 1992 autobiographical essay in *Annual Review of Biophysics*. The technical results were published in *PNAS* in 1958.

Myth Risk Low. While the "lone scientist at night" is a dramatic trope, the technical specificity of the 1:1 ratio and the 100% activity restoration are documented in the original 1958 paper and subsequent peer-reviewed retrospectives.

SCENE #2: The Oxford Corridor and "Fred's Folly"

Summary

During a sabbatical at Oxford, Richards confronted the "visualization bottleneck" of protein crystallography. The raw data of the field—electron density maps—consisted of stacks of two-dimensional contour plots that were nearly impossible to interpret as three-dimensional shapes. Richards utilized his engineering background to build an optical device that allowed scientists to "see" their physical models inside the data.

Date/Place 1967–1968; Laboratory of Molecular Biophysics, Oxford University, UK (specifically the hallway outside David Phillips' office).

People Frederic Richards; David Phillips (host); a young Thomas Steitz (postdoc); and Chris Anfinsen (visitor).

Concrete Details The device, known formally as an "optical comparator" but colloquially as the "Richards Box" or "Fred's Folly," was a massive wooden and glass structure. Because the laboratory was overcrowded, Richards assembled the box in the hallway, where it became a landmark for passersby. The apparatus used a half-silvered mirror set at a 45-degree angle. On one side of the mirror was a stack of transparent acetate sheets (Mylar) printed with electron density contours. On the other side was a brass "Kendrew-style" wire model of the protein. When a researcher looked through the mirror, the reflection of the brass atoms appeared to be floating precisely within the glowing contours of the density map, allowing for rapid, intuitive model building.

Why it Matters Before the age of computer graphics, the Richards Box was the global standard for protein structure determination. It was used to solve the first all-atom model of Ribonuclease S and was a precursor to modern molecular visualization software. It represents Richards' "Yankee" philosophy of building the tools necessary to solve the science.

Evidence Quality High. Detailed schematic diagrams were published in the *Journal of Molecular Biology* (1968). Numerous photographs exist of the box in the Oxford hallway.

Myth Risk Medium. Some accounts suggest Richards "invented" the idea of optical comparators; however, similar devices existed in other engineering fields. His contribution was the specific application to the 3.5 Å resolution problems of crystallography.

SCENE #3: The Beachfront Mutiny (The PDB Petition)

Summary

In the early 1970s, as the first protein structures were being solved, the community was divided over whether the atomic coordinates should be public or private. Richards led a movement to mandate that structure data be deposited into a central repository, essentially inventing the concept of "open data" in biology.

Date/Place June 1971; Cold Spring Harbor Laboratory (CSHL), New York.

People Frederic Richards; Michael Rossmann; Helen Berman; and Walter Hamilton.

Concrete Details The debate took place during a pivotal meeting on "Structure and Function of Proteins at the Three-Dimensional Level". The atmosphere was described as "intense and exciting," with discussions moving from the lecture hall to the CSHL lawn and the beach. Richards, frustrated by the reluctance of some labs to share their hard-won coordinates, drafted a petition. He was described by colleagues as an "outspoken critic" of those who would "trivialize achievement" by withholding data. He spent the meeting lobbying fellow scientists to sign a document that would eventually be sent to major journals, demanding that they make coordinate deposition a requirement for publication.

Why it Matters This scene marks the birth of the Protein Data Bank (PDB). Richards' insistence on transparency ensured that structural biology would be a collaborative global enterprise rather than a collection of secretive laboratories. This data-sharing culture directly enabled the eventual development of modern computational biology and AI-driven structure prediction (like AlphaFold).

Evidence Quality High. The founding of the PDB is well-documented in historical reviews by Helen Berman and others. The "Richards committee" is a recognized entity in the history of the International Union of Crystallography.

Myth Risk Low. The existence of the formal petition and the subsequent policy changes by journals like *Nature* and *JBC* are matter-of-fact historical records.

SCENE #4: The Transatlantic Voyage of Hekla

Summary

Richards' commitment to his personal life and his refusal to be consumed by the "social forces that erode time" were manifested in his decision to travel to his Oxford sabbatical by captaining his own sailboat across the Atlantic Ocean.

Date/Place Summer 1967; Atlantic Ocean (U.S. to UK).

People Frederic Richards (Captain); his wife Sally; and a small crew.

Concrete Details Richards sailed a 40-foot vessel named *Hekla*. This was not a leisure cruise but a grueling physical challenge requiring 24-hour watches, celestial navigation (pre-GPS), and the management of heavy weather. Richards believed that this type of total disconnection from the laboratory was essential for scientific creativity. His colleagues noted that he would return from such excursions "refreshed and ready to work". He performed this crossing twice in his life, cementing his reputation as a "charismatic, fun-loving" leader who practiced what he preached about work-life balance.

Why it Matters The voyage illustrates Richards' "Yankee self-reliance" and the "sheer joy of doing" that he brought to his science. It is the primary anecdote used by colleagues to explain his unique leadership style at Yale—one that valued the "full life" as a prerequisite for great thinking.

Evidence Quality High. Mentioned in every major biographical memoir (NAS, Science, Nature) and frequently recounted by his successor, Thomas Steitz.

Myth Risk Low. The boat names and the fact of the voyages are well-documented, though specific dramatic events during the crossing (e.g., specific storms) are largely kept within family oral tradition.

SCENE #5: The High School Cannonball Experiment

Summary

Before he was a world-famous biochemist, Richards was a student at Phillips Exeter Academy who was given the "unsupervised run of the laboratories." He attempted a massive physical experiment to measure the mass of the Earth.

Date/Place 1943; Phillips Exeter Academy, Exeter, New Hampshire.

People Frederic Richards and his roommate, John King.

Concrete Details The two students attempted to replicate the 1797 Cavendish experiment to measure the universal gravitational constant (G). The apparatus was home-built and featured two 100-pound cannonballs as the attracting masses. Richards recalled that the school's science department permitted students to work "unsupervised... outside of class hours," an attitude that he claimed "played a strong role... in cementing our commitment to scientific careers". The experiment required extreme precision to measure the minute gravitational pull

between the weights, and although the results were not of publication quality, the process of building the instrument was the real achievement.

Why it Matters This scene highlights the early origin of Richards' "hands-on" experimental philosophy. It also illustrates the institutional trust that Richards later tried to replicate in his own department at Yale, where he encouraged postdocs and students to pursue "fun" and "ingenious" projects.

Evidence Quality High. Richards recounted this in his *Annual Review* autobiography, and photographs of the high school apparatus were preserved and published in his memorial pieces.

Myth Risk Low. The detail of the "100-pound cannonballs" is specific and documented in his own reminiscences.

SCENE #6: The Fiber-Glassing Party (Sally's Baage)

Summary

In his later years, Richards applied his engineering skills to build a specialized utility boat for Long Island Sound. Rather than making it a private hobby, he turned the construction into a collaborative event for the local scientific community.

Date/Place 1970s–1980s; Richards' home and backyard "Fix-it Shop" in Guilford, Connecticut.

People Frederic Richards; his wife Sally; and a rotating cast of "friends and scientists" from Yale.

Concrete Details The boat was a utility derrick-style barge, built from plans by naval architect Ron Noe. Richards named it "Sally's Baage," using the phonetic spelling of her Maine accent. The "Fix-it Shop" in his backyard was filled with "fascinating gadgets" and tools. For the final construction phase, Richards hosted a day-long "fiberglassing party," where colleagues and students applied layers of resin and glass cloth to the hull. He was seen working alongside his juniors, covered in resin, demonstrating that the "Sterling Professor" was still a bench scientist at heart.

Why it Matters This scene serves as an allegory for Richards' leadership of the Yale MB&B department. He built the department (the "barge") from the ground up, hiring giants like Tom Steitz and Joan Steitz, and he did so by fostering a culture where everyone helped with the "manual labor" of science.

Evidence Quality Medium. This comes from the personal recollections of B. K. Lee in the Yale obituary.

Myth Risk Low. The phonetic name "Baage" is a detail unlikely to be invented and is consistent with Richards' documented sense of humor.

B) DIRECT QUOTES FROM THE SCIENTIST

"This discovery came as a surprise to the scientific community at that time.... In retrospect, this may have been the high point of my career in terms of excitement."

- **Context/Date**: Reflecting on the 1957 Ribonuclease S experiment.
- **Source**: Richards, F.M. (1992). "Whatever happened to the fun? An autobiographical investigation." *Annual Review of Biophysics and Biomolecular Structure*, 21: 1–22.

"Protein crystallography is rather like a dog chasing cars. Both are probably foolish and certainly heroic endeavors. But in both cases, the real problem is: suppose you actually manage to catch one, what in the world do you do with it?"

- **Context/Date**: A lecture describing the challenge of interpreting structural data after the technical hurdles of crystallography are cleared.
- **Source**: Quoted in Richardson, D. C. & Richardson, J. S. (2006). "The Richardson Perspective" and various archival structural biology lectures.

"In summary it seems reasonable that there is a generalized driving force with no particular specificity produced by the tendency of non-polar groups to minimize their interaction with water."

- **Context/Date**: From his 1977 definitive review on protein packing and the hydrophobic effect.
- **Source**: Richards, F.M. (1977). "Areas, volumes, packing and protein structure." *Annual Review of Biophysics and Bioengineering*, 6: 151–176.

"Looking, and especially seeing, is an active process of interpreting, of finding and emphasizing patterns, of adding things and deleting things."

- **Context/Date**: Discussing the philosophy behind the "Richards Box" and model-building.
- **Source**: Archival records of the Yale Center for Structural Biology, reprinted in memorial retrospectives.

"Whatever happened to the fun?"

- **Context/Date**: The opening line and title of his 1992 autobiography, critiquing the bureaucratic nature of modern academic science.

- **Source**: Richards, F.M. (1992). "Whatever happened to the fun? An autobiographical investigation." *Annual Review of Biophysics and Biomolecular Structure*, 21: 1–22.

C) HUMAN REALITY DETAILS

- **1600s Heritage**: Richards was born into an "old New England" family that had settled in Connecticut in the 1600s, a background that influenced his reserved but direct "Yankee" personality — Source: http://biographicalmemoirs.org/pdfs/richards-frederic.pdf & https://limlab.ucsf.edu/pdfs/lim_2009.pdf
- **The "Chief" Persona**: At Yale, Richards was universally referred to as "The Chief," a nickname that reflected both his leadership and the respect he commanded from Nobel laureates and first-year students alike — Source: https://limlab.ucsf.edu/pdfs/lim_2009.pdf
- **Glassblowing Expertise**: He learned glassblowing in high school and used this skill throughout his career to build his own lab equipment, including a specialized micro-balance for his PhD thesis — Source: http://biographicalmemoirs.org/pdfs/richards-frederic.pdf
- **The 4 PM Routine**: Colleagues noted that Richards had a disciplined work habit: at "4 PM sharp," he would leave his office and spend exactly 1–2 hours at the Yale computing center writing and submitting programs for an IBM 360 — Source: https://pmc.ncbi.nlm.nih.gov/articles/PMC2762580/
- **Ice Hockey Passion**: Even as an emeritus professor in his 70s, Richards was an active ice hockey player, often bringing the same "square-jawed directness" to the rink as he did to the lab — Source: https://limlab.ucsf.edu/pdfs/lim_2009.pdf
- **Marianna the Role Model**: His interest in chemistry was sparked by his older sister, Marianna, who was pursuing a PhD in biochemistry when he was a child; she provided him with his first chemistry sets and "glassware" — Source: http://biographicalmemoirs.org/pdfs/richards-frederic.pdf
- **Mentorship Style**: He was known for being "always direct and a lot of fun to be around," famously treating a "loyal technician" with the same level of intellectual respect as a "Nobel laureate" — Source: https://mbb.yale.edu/sites/default/files/files/Richards-Obit.pdf & https://limlab.ucsf.edu/pdfs/lim_2009.pdf
- **The "Dozing" Seminar Trick**: Richards famously appeared to doze off during research seminars at Yale, only to "awaken" at the end and ask a question that perfectly identified the heart of the speaker's problem — Source: https://limlab.ucsf.edu/pdfs/lim_2009.pdf
- **Work-Life Separation**: He strictly reserved 4–6 weeks every summer for sailing, believing that "at least a month off" was mandatory to return "refreshed and ready to work" — Source: https://mbb.yale.edu/sites/default/files/files/Richards-Obit.pdf & http://biographicalmemoirs.org/pdfs/richards-frederic.pdf
- **Critique of Bureaucracy**: He was an outspoken critic of social forces that "erode time and trivialize achievement," often lamenting that scientists spent too much time on administration and not enough on "the sheer joy of doing" — Source: https://mbb.yale.edu/sites/default/files/files/Richards-Obit.pdf &

http://biographicalmemoirs.org/pdfs/richards-frederic.pdf

D) SCIENCE CONNECTION (Intro Biochemistry Bridge)

Best-fit Intro Biochemistry Topic: Protein Structure and Folding

The work of Frederic Richards is the primary experimental evidence for how **non-covalent interactions** (the hydrophobic effect, hydrogen bonds, and van der Waals forces) stabilize a protein's tertiary structure.

Precise Contribution

Richards discovered that the enzymatic activity of Ribonuclease A could be fully restored by the non-covalent association of two inactive fragments (S-peptide and S-protein), proving that the structural "fold" of a protein is held together by its internal packing rather than just its covalent backbone.

Teachable Bridge Explanation

In introductory biochemistry, students often view a protein as a "string" of beads (amino acids) that must stay connected to function. Richards showed this wasn't strictly true. He used a molecular "scissors" (subtilisin) to cut the string of the Ribonuclease enzyme into two separate pieces. Neither piece could work alone. However, when he simply put them back in the same test tube, they "found" each other and stuck together so perfectly that the enzyme started working again at 100% efficiency. This demonstrates that the **Hydrophobic Effect**—the tendency of "oily" amino acids to hide from water—creates a force strong enough to pull a protein into its functional shape even if its backbone is broken.

Scientist’s Own Analogy

Richards compared the interior of a protein to a **three-dimensional jigsaw puzzle**. He argued that the core of a protein is not a "loose gel" but is packed as tightly as an organic crystal (like a hard block of sugar), with the amino acid side chains fitting together with "exquisite complementarity".

E) COMPLICATIONS, DISPUTES, OR CONTROVERSIES

Item: The PDB Coordinate Sharing "War"

- **What’s Agreed**: In the early 1970s, there was no centralized system for sharing the atomic coordinates (x, y, z positions) of solved proteins. Most researchers exchanged data through personal letters and punched cards.
- **What’s Disputed**: Many prominent crystallographers (the "opponents") felt that coordinates were intellectual property that should be kept secret until the original lab had

"exhausted" all potential publications from the data. Richards led an aggressive "mutiny" of younger scientists who argued that public funding required public data.
- **Sources**: Berman, H. M. (2008) "The Protein Data Bank: a historical perspective" ; Richards' own petition from the 1971 CSHL meeting.

Item: The "Failure" of Myoglobin Packing Predictions

- **What's Agreed**: Richards attempted to use early computers to "predict" how a protein would fold by simulating the packing of helices in myoglobin.
- **What's Disputed**: Richards later admitted that these specific computational experiments did *not* lead to a "flash of illumination" or a breakthrough in the folding problem.
- **Significance**: While Richards viewed this as a personal setback, the "packing density" algorithms he wrote for this "failed" project became the foundation for almost all modern drug-design software.
- **Sources**: Baldwin, R. L. (NAS Biographical Memoir) ; Richards, F. M. (1992 Autobiography).

F) "NOT ON WIKIPEDIA" GOLD

1. **The "Derrick" Barge Construction**: Richards built a utility boat ("Sally's Baage") equipped with a derrick (a small crane) to explore Long Island Sound. He didn't just sail for leisure; he built a specialized engineering platform for the water — Why it's distinctive: Highlights his dual identity as a "Yankee Engineer" and a scientist — Source: https://mbb.yale.edu/sites/default/files/files/Richards-Obit.pdf
2. **The 1957 Chronology**: Richards was so precise about his "Eureka" moment that he recorded the exact time (7:00 PM) and the specific instrument used (Beckman-DU), allowing historians to reconstruct the evening in a way few other discoveries are documented — Why it's distinctive: Rare precision in scientific biography — Source: https://mbb.yale.edu/sites/default/files/files/Richards-Obit.pdf
3. **The "Fix-it Shop" Backyard**: Richards' Guilford home featured a detached building known as the "Fix-it Shop," which was more famous among his Yale colleagues than his official office. It was where the "Richards Box" was prototyped — Why it's distinctive: Anchors his scientific creativity in physical craft — Source: https://mbb.yale.edu/sites/default/files/files/Richards-Obit.pdf
4. **Hiring the "Steitz Dynasty"**: Richards was responsible for recruiting Thomas Steitz, Joan Steitz, Peter Moore, and Don Engelman to Yale in a single era, effectively creating the most successful structural biology department in history — Why it's distinctive: Demonstrates his "eye for talent" and institutional vision — Source: https://tomsteitzsymposium.yale.edu/news/retrospective-asbmb & https://news.yale.edu/2009/10/07/quotes-and-about-yale-s-nobel-prize-winner-thomas-steitz
5. **The 100-Pound Cannonballs**: In 1943, as a high schooler, he almost caused a structural failure in a school building by suspending 100-pound cannonballs to measure

gravity — Why it's distinctive: Shows a fearless (and slightly dangerous) early commitment to physical data — Source: https://limlab.ucsf.edu/pdfs/lim_2009.pdf

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G) DATA TABLES FOR ARCHIVAL ANALYSIS

Table 1: Comparison of Ribonuclease A vs. Ribonuclease S

Property	Ribonuclease A (Native)	Ribonuclease S (Richards' Version)
Covalent Chain	Continuous (1-124 residues)	Split (1-20 and 21-124)
Linkage between parts	Peptide Bond (Covalent)	Non-covalent (Hydrophobic/H-bonds)
Enzymatic Activity	100%	100% (When mixed 1:1)
Stability	High	Labile (Dissociates in urea)
Source of Discovery	Natural (Pancreas)	Engineered (Subtilisin digestion)

Table 2: Chronology of Major Institutional Hires (Yale MB&B)

Year	Faculty Member	Research Area	Recognition
1967	Harold Wyckoff	X-ray Crystallography	Co-solver of RNase S
1970	Thomas Steitz	Ribosome Structure	Nobel Prize 2009

1970	Joan Steitz	RNA Processing	Lasker Award / NAS
1971	Peter Moore	Neutron Scattering	NAS Member
1971	Donald Engelman	Membrane Biophysics	NAS Member

H) FLAGGED “MYTH-RISK” DETAILS

- **Claim**: Richards "stumbled" upon RNase S when a sample of Ribonuclease was contaminated by subtilisin-producing bacteria.
    - **Where it Appears**: Various informal student blog posts and "history of science" trivia sites.
    - **Why Sourcing is Weak**: Richards’ own memoirs state the experiment was a deliberate follow-up to Sumner Kalman’s work at the Carlsberg Lab.
    - **Best Available Evidence**: The 1955 *Biochimica et Biophysica Acta* paper clearly lists "Degradation of ribonuclease by subtilisin" as a planned study.
- **Claim**: The Richards Box was called "Fred's Folly" because it didn't work.
    - **Where it Appears**: Oral traditions at Oxford.
    - **Why Sourcing is Weak**: The device worked perfectly and was used for a decade. The "Folly" name referred to its massive, hallway-clogging size, not its utility.
    - **Best Available Evidence**: The completion of the 2.0 Å RNase S structure in 1970 using the device.

LENS RECOMMENDATION

Lens Scoring Rubric (0–3 each)

Lens	Score	Justification
Eureka	3	The Dec 7, 1957 titration is a perfectly documented "needle-moving" moment with high texture (Beckman-DU, 7:00 PM).

Failure	2	The myoglobin computer simulation project was a well-documented "non-breakthrough" that Richards analyzed in his 1992 memoir.
Mentor	3	His nickname "The Chief" and the success of his "children" (Steitz, Moore, etc.) are central to his historical legacy.
Controversy	3	The PDB petition is a high-stakes conflict over the "soul" of science (private vs. public data).
Human Detail	3	The "Yankee Engineer" archetype—building boats, blowing glass, the Richards Box—is the most distinctive aspect of his character.
Last Experiment	2	He was still doing bench work in his 70s, but there is no single dramatic "final" scene equivalent to the 1957 titration.

Primary Recommended Lens: Eureka (The Restoration of Life)

- **Why it fits**: This lens leverages the strongest, most vividly documented scene in the collection (Scene #1). It allows the student to write a narrative that starts with the physical "silence" of the inactive fragments and ends with the "rising needle" of the spectrophotometer. This scene also provides the most direct connection to the BIOL 3030 curriculum (non-covalent bonds).
- **Best opening scene**: Scene #1 (The 7:00 PM Titration).
- **One quote to use early**: Quote #1 ("This discovery came as a surprise... the high point of my career.").
- **Cautions**: Ensure the student understands that the "surprise" was scientific, not accidental; it proved the *forces* of folding were non-covalent.

Backup Lens: Human Detail (The Yankee Engineer)

- **Why it fits**: If the student prefers a more "character-study" approach, the lens of the "Engineer who built his way to the top" is supported by Scene #2 (The Richards Box) and Scene #5 (The Cannonballs).

- **Best opening scene**: Scene #2 (Assembling the "Folly" in the Oxford hallway).
- **One quote to use early**: Quote #2 (The "dog chasing cars" analogy for crystallography).
- **Cautions**: Must be careful not to make him sound like *only* an engineer; the goal is to show how engineering *served* the biochemistry.